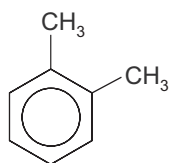
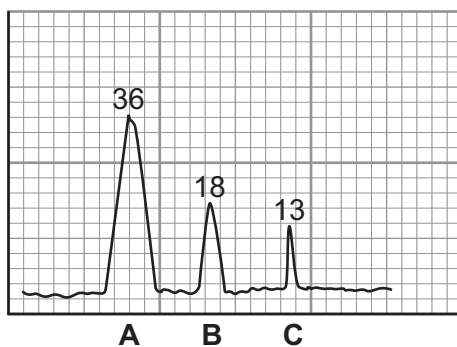


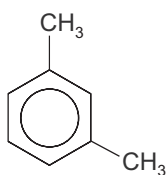
4. State the reagent used to produce phenylmethanol from benzaldehyde. [1]

.....

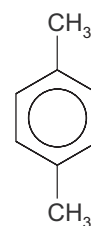
5. The gas chromatogram shows peaks for the products formed by the Friedel-Crafts alkylation of methylbenzene at 0 °C.



A



B



C

Use the data to calculate, to the appropriate number of significant figures, the percentage of 1,3-dimethylbenzene, which is present in the products. [1]

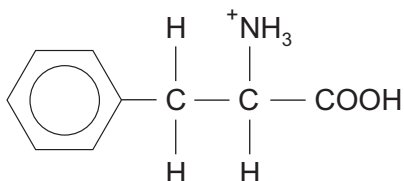
Percentage = %

6. Give the systematic name of a compound of formula $C_5H_{10}O_2$ that gives propan-1-ol as one of the products when it is warmed with aqueous sodium hydroxide. [1]

.....



4. (a) In acid solution the formula of the ion formed from phenylalanine is



Give the formula of the ion formed from phenylalanine in basic solution.

[1]

- (b) The melting temperatures of four compounds are shown in the table.

Compound	Melting temperature / °C
$\text{CH}_3\text{CH}_2\text{COOH}$	-21
$\text{CH}_3\text{CH}(\text{OH})\text{COOH}$	18
$\text{CH}_3\text{CHBrCOOH}$	26
$\text{CH}_3\text{CH}(\text{NH}_2)\text{COOH}$	289

Explain why the melting temperature of 2-aminopropanoic acid is very much higher than the other acids in the table.

[2]

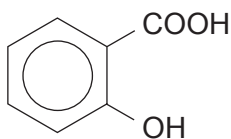
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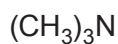
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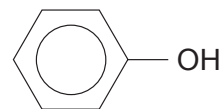
4. Using the letters provided, place these three compounds in order of **increasing** acidity. [1]



A



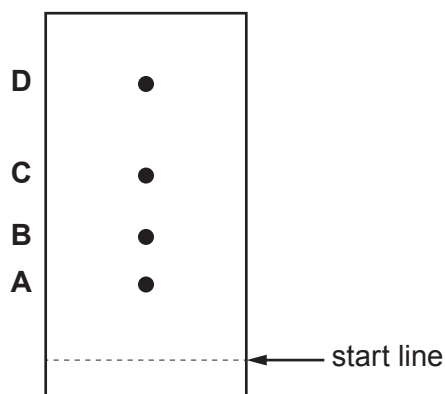
B



C

least
acidicmost
acidic

5. A student produced the following chromatogram of a mixture of four amino acids. The student forgot to mark the position of the solvent front.



The R_f values of the four amino acids are as follows.

Amino acid	R_f value
alanine	0.59
glycine	0.42
leucine	0.88
serine	0.36

State and explain which of the spots is likely to be alanine. [1]

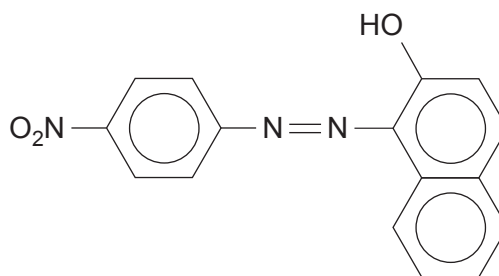
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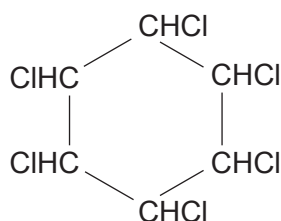
4. The formula of the azo dye, Para Red, is shown below.



Give the structure of the compound that couples with 4-nitrobenzenediazonium chloride to give this dye.

[1]

5. Hexachlorocyclohexane can be used as an insecticide.



Deduce the empirical formula of this compound.

[1]

6. Give the structure of the organic product of the reaction below.

[1]

